

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Smart Complaint Management System

Campus Facility Complaint Registration, Routing and Resolution Tracking

SYNOPSIS / PROJECT PROPOSAL

(Deliverable 1)

Item	Details
Project Title	Smart Complaint Management System
Group Number	Team 6
Course	UE24CS341A : Software Engineering
Development Methodology	Agile
Version	1.0
Date	2 September 2026

Team Profile

Sl. No.	SRN	PRN	Student Name
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1. Proposed Project Description

The Smart Complaint Management System (SCMS) is a web-based complaint registration and resolution tracking system for campus and hostel facility complaints at PES University. At present, complaints about hostel rooms, water supply, electrical faults, cleaning, furniture and other campus facilities are raised through registers kept at the facility office, WhatsApp messages, phone calls, verbal requests to wardens, and repeated visits to the administration office. Because the complaint is not recorded in one place, the person who raised it has no reliable way of knowing whether it was accepted, who is working on it, or when it will be attended to.

SCMS brings the complete complaint cycle into one portal. A student, hostel resident or staff member signs in, submits a complaint with a category, description, location and supporting photographs, and receives a unique complaint number. The system applies configured routing rules to send the complaint to the correct department queue, the department head assigns it to a resolver, and the complainant can follow the status at every step. Each complaint has a due date based on its category, and a complaint that is not resolved within that period is escalated automatically to the department head. The complainant can give feedback after resolution and reopen the complaint if the problem persists.

The system also supports complaints that a person may not wish to raise under their own name. An anonymous complaint can be submitted without login and tracked later using a tracking reference. Departments and administrators get dashboards and reports showing pending complaints, overdue complaints and average resolution time. The scope of every feature is kept direct and limited so that the team can build, test, document and demonstrate the complete system within the course schedule.

2. Problem Statement

The main problem is that campus and hostel facility complaints are recorded and followed up informally, so no single record of a complaint exists. A complaint may be written in a register at one office, sent as a message to a warden, or conveyed verbally to maintenance staff. The complainant cannot check the status without contacting the office again, and there is no acknowledgement or reference number to quote. Complaints raised on holidays or after office hours may not be recorded at all.

The same gap creates avoidable work for the facility staff. Complaints reach the wrong department and have to be redirected manually, the same complaint is reported by several residents and worked on more than once, and there is no record of who was assigned a complaint or when it was closed. Nothing marks a complaint that has been pending beyond a reasonable time, so an overdue complaint stays unnoticed until the complainant follows up in person. Residents who are uncomfortable reporting an issue under their own name, such as a complaint against a facility or its handling, have no safe way to report it.

Administration also has no consolidated view. There is no reliable way to answer how many complaints are pending, which department is slow, which category of complaint recurs most, or how long resolution typically takes, because the information needed to answer these questions was never captured in a structured form. This affects students and hostel residents who depend on timely repairs, facility staff who handle the work, and the administration office that is accountable for it.

3. Target Users and Stakeholders

User / Stakeholder	Role in the System
Student / Hostel Resident (Complainant)	Registers complaints with description and photographs, tracks the status of own complaints, gives feedback after resolution, and reopens a complaint if the issue persists.
Anonymous Complainant	Submits a complaint without signing in and checks its status later using the tracking reference issued at submission.
Department Staff / Resolver	Views complaints assigned to them, updates work progress, records the resolution action taken, and marks the complaint resolved.
Department Head / Warden	Reviews the department queue, assigns and reassigns complaints to resolvers, handles escalated and overdue complaints, and monitors department performance.
Administrator	Manages user accounts and roles, departments, complaint categories, routing rules and resolution timelines, and views workspace-wide reports.
Campus Facility and Hostel Administration Office	Institutional stakeholder that uses the consolidated reports on pending, overdue and resolved complaints for oversight and planning.

4. Objectives

- Provide a single portal through which campus and hostel facility complaints are registered, acknowledged with a unique complaint number, and tracked to closure.
- Allow complaints to be supported by photographs or documents so that the reported issue can be understood without a physical inspection first.
- Route every complaint automatically to the responsible department using configured category and location rules, instead of manual redirection.
- Track a resolution timeline for every complaint and escalate a complaint that crosses its due date to the department head without manual intervention.
- Keep the complainant informed at submission, assignment, resolution and escalation through in-app and email notifications.
- Allow complaints to be raised anonymously and tracked without disclosing the identity of the complainant.
- Give the administration office reports on pending, overdue and resolved complaints, department load and average resolution time.
- Build the system in short Agile iterations in which each team member owns one complete functional area end to end.

5. Scope

SCMS will cover the complaint activities required for a complete first working version. The scope is deliberately limited so that every selected function can be built, tested and demonstrated by the team.

Complainant access: Students, hostel residents and staff sign in, register complaints with category, description, location and attachments, and view the status and history of their own complaints only.

Anonymous submission: A complaint can be submitted without an account. The system issues a tracking reference which is the only way to view that complaint's status, and the complainant's identity is not stored.

Routing and assignment: Routing rules maintained by the administrator map a complaint category and location to a department queue. The department head assigns the complaint to a resolver and may reassign it.

Resolution lifecycle: A complaint moves through submitted, assigned, in progress, resolved and closed, and may be reopened or rejected with a recorded reason. Every status change is recorded with the user and time.

Timelines and escalation: Each category carries a resolution period. A scheduled check marks complaints that cross the period as escalated and raises the escalation level for the department head.

Notifications and feedback: In-app and email notifications are sent on submission, assignment, resolution and escalation. The complainant can rate the resolution and reopen the complaint within the permitted period.

Administration and reporting: The administrator maintains users, roles, departments, categories, routing rules and timelines. Reports cover complaint counts by status, department and category, overdue complaints and average resolution time.

Out of scope: The system will not integrate with university ERP, hostel allotment or payroll systems, and will not perform any financial or billing transaction. It will not use machine learning, image recognition or predictive analysis for complaint handling. There will be no native mobile application, no SMS or WhatsApp channel, and no offline mode. Inventory, purchase and work-order costing for repairs are not included, and the system is limited to a single campus with an English interface.

6. Functional Features

Feature	Planned Functionality
Registration and login	Users register or are created by the administrator and sign in with a password. The signed-in role decides the functions available to the user.
Role and access control	The administrator maintains roles, departments and staff membership, so each user can act only on the complaints permitted for that role.

Feature	Planned Functionality
Complaint registration	The complainant submits category, description, location and hostel or block details, and the system issues a unique complaint number in the format SCMS-YYYY-NNNNNN.
Evidence attachments	Up to three photographs or documents in JPEG, PNG or PDF form, each within the configured size limit, can be attached to a complaint as supporting evidence.
Anonymous complaint	A complaint can be submitted without login. The system returns a tracking reference which the complainant uses later to view status without revealing identity.
Automatic routing	The system applies administrator-defined rules on category and location to place the complaint in the correct department queue at submission.
Assignment and reassignment	The department head views the department queue and assigns a complaint to a resolver, with reassignment recorded in the complaint history.
Timeline and escalation	Each complaint carries a due date from its category. A scheduled check escalates complaints that cross the due date to the department head and records the escalation level.
Status tracking	The complaint moves through submitted, assigned, in progress, resolved and closed, with reopen and reject paths. The complainant sees the current status and the full history.
Notifications	In-app and email notifications are generated on submission, assignment, resolution and escalation for the complainant and the concerned staff.
Feedback and reopen	After resolution the complainant records a rating and remarks, or reopens the complaint within the permitted period and limit. Complaints not reopened are closed automatically.
Dashboards and reports	Department heads and the administrator view complaint counts by status, department and category, overdue complaints and average resolution time, with export for the administration office.

7. Team-wise Functional Ownership

The work is divided into four roles that follow the path of a complaint through the system, so that the output of one role becomes the input of the next. Each member owns one complete functional area, including its screens, data handling, validation, unit tests and documentation. All four members take part in requirements discussion, design reviews, Git commits, code review, integration and the final demonstration.

Member and Role	Primary Functional Ownership	Handover to the Next Stage
Anushka K Kishore (PES1UG24CS074): Foundation and Access	Project and database setup; registration and login; password rules, lockout and session timeout; role definitions and role-based access enforcement; department and staff membership records; administrator screens; administrator reports and export	Working authentication, the four user roles, department records and the base page template
Atul Suresh (PES1UG24CS098): Complaint Intake and Evidence	Complaint submission with category, description and location; input validation; complaint number generation; attachment upload, validation and viewing; anonymous submission with tracking reference lookup and request throttling	A stored complaint record with attachments and a unique complaint number
Chaitanya Krishna Makkena (PES1UG24CS122): Routing, Assignment and Escalation	Complaint categories with resolution periods; routing rule maintenance; automatic routing on submission; department queues; assignment and reassignment by the department head; due-date calculation; scheduled overdue check and automatic escalation	A complaint placed in the correct department queue, assigned to a resolver, with a due date and escalation state
Tanmay Sridhar (PES1UG25CS849): Resolution, Notifications and Tracking	Status transition engine and complaint history; resolver progress and resolution notes; reject, reopen and auto-close paths; in-app and email notifications; feedback and rating; complainant and department head dashboards	A closed complaint with full history and feedback, completing the cycle back to the complainant

Each member also carries one shared responsibility for the whole team: Anushka the security validation of access rules, Atul the test framework and coverage reporting along with demonstration data, Chaitanya the continuous integration pipeline, and Tanmay the repository, branch policy, product backlog and documentation upkeep.

8. Plan of Work

The sprints follow the same order as the roles. Anushka delivers the foundation first, Atul then creates the complaint record, Chaitanya routes and assigns it, and Tanmay takes it to closure. A member with no feature work in an early sprint takes up the shared responsibilities and pairs with the member who is active.

Sprint	Planned Output	Acceptance Evidence
Sprint 1	Project and database setup, registration and login, role definitions and access enforcement, base page template, repository and backlog, test framework and continuous integration pipeline	User stories in the backlog, commits from all four members, working sign-in with role-based landing pages, first pipeline run
Sprint 2	Department and staff membership records, administrator screens, agreed complaint record fields, complaint submission with validation, complaint number generation and attachment upload	Complaint created and listed with a valid number and attachments, department records maintained, unit tests for validation rules
Sprint 3	Password rules, lockout and session timeout, attachment viewing, anonymous submission with tracking lookup, complaint categories and routing rules, automatic routing, department queues and assignment	Complaint routed to the correct department and assigned to a resolver, anonymous tracking demonstrated, access rules verified
Sprint 4	Reassignment with reason, due-date calculation, status transition engine, resolver progress updates, complaint history and reject path, throttling on anonymous lookup	Complete walkthrough from submission to resolved with history entries, reassignment recorded, test results for the transition rules
Sprint 5	Scheduled overdue check and automatic escalation, in-app and email notifications, reopen with limit, auto-close and feedback capture	Escalation raised on an overdue complaint, notification and email records, feedback captured against a closed complaint
Sprint 6	Complainant and department head dashboards, administrator reports and export, security validation pass, demonstration data, documentation, defect correction and final system demonstration	Reports matching sample data, role-access and security checks, stable build, coverage and test summary, team demonstration

9. Proposed Technology and Tools

The allocated technology for this project is Python (Flask or Django) or PHP (Laravel). Django is selected because the system is built around authenticated users with distinct roles, records that move through a defined status flow, and administrator-maintained master data. Django provides user authentication, permission checks, an administration interface and an ORM as part of the framework, so the team can spend its effort on the complaint workflow rather than on rebuilding these basics.

Purpose	Selected Technology / Tool
Web application framework	Python with Django
User interface	Django templates with HTML, CSS and JavaScript
Database	PostgreSQL for deployment, SQLite during development

Purpose	Selected Technology / Tool
Authentication and roles	Django authentication with role-based permission checks
Attachments	Django file storage with file type and size validation
Email notifications	Django email backend over institutional SMTP
Scheduled escalation check	Django management command run on a schedule
Version control	Git and GitHub
Agile planning	GitHub Projects for the product backlog and sprint board
Testing	pytest with the Django test client and coverage reporting
Continuous integration	Jenkins pipeline for build and test on each push
Development environment	Standard code editor and modern web browser